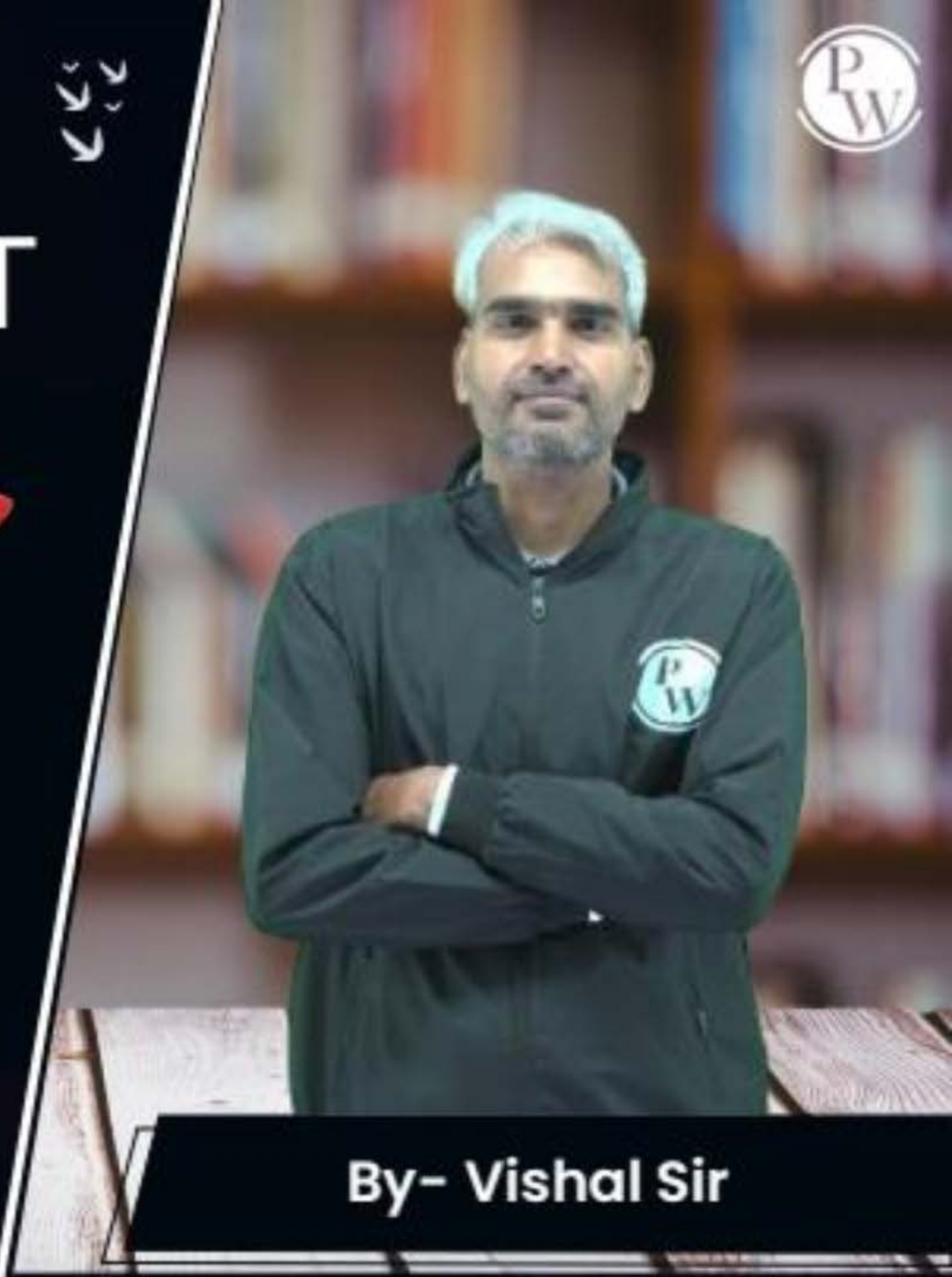
Computer Science & IT



Database Management System

Query Languages

Lecture No. 09



By- Vishal Sir

Recap of Previous Lecture



Topic Comparison with regular expression

Topic Practice questions on SQL

Slide

Topics to be Covered



Topic Tuple relational calculus (TRC)



Slide



Topic : Relational Calculus

- Relational Calculus is a non-procedural query language.
- It uses predicate calculus(First Order Logic) instead of algebra.

Non-procedural

- Relational calculus provides description about the query to get the result, whereas Relational Algebra gives the method to get the result.

Procedural



Topic : Types of Relational Calculus

There are two types of Relational Calculus.

1. Tuple Relational Calculus: Works on tuple (or rows)
2. Domain Relational Calculus: Works on Domain of attribute (or column)



Topic : Syntax of TRC

Basic syntax of TRC is { t | P(t) }

where t is a tuple variable and P(t) is a predicate formula that describes the conditions that must be satisfied by the resulting tuples.

Student

Sid	Sname	Marks
S_1	A	50
S_2	A	65
S_3	B	72

1. It results the set of tuples t, s.t. P(t) is true for t.
2. t ∈ *Relation* denotes that the t is a tuple from Relation, also denoted using Relation(t). $t \in Relation \equiv Relation(t)$
3. If t is a tuple variable, "t[A]" / "t.A" is value of attribute A in tuple t.
4. P is a predicate formula similar to that of the predicate calculus.



Topic : Equivalence between SQL, RA and TRC

- SELECT distinct * FROM Employee :- Produces Complete employee table (* → Every thing)
- (Employee) : Produces Complete Employee table
- {t | t ∈ Employee}
  or
  {t | Employee(t)}

Predicate Formula

{ t | t ∈ Employee }

Predicate Formula

{ t | P(t) }

tuple Variabl.

Predicate formula

{ for all the tuples of Employee table this Predicate Formula will be true
∴ All tuples of Employee table will be selected.



Topic : Free and Bounded variables

- A variable which is preceded by a quantifier is called bounded variable.

eg: $\exists S \in Student$

S is preceded by $\exists$, $\therefore$ S is a bounded Variable

- A variable which is not preceded by any quantifier is called free variable.

- There is an important restriction:

  The variable 't' that appears before '|' symbol must be the only free variable in formula P(t).

  In other words all other variables must be bounded using quantifier

In $\{ t \mid P(t) \}$

this "t" is called output tuple Variable, it is used to produce the output.



Topic : Equivalence between SQL, RA and TRC

SQL

- SELECT distinct ename FROM Employee

- π_{ename} (Employee)

- { t.ename | t ∈ Employee}

  ||| or

  { t[ename] | t ∈ Employee}

to produce only ename attribute from the selected tuples

't' is the output variable

or

{ t | ∃S ∈ Employee (t[ename]= S[ename])}

or

{ t | ∃S ∈ Employee (t.ename= S.ename)}

tuple from Employee is selected.

Creation of the attributes in the o/p tuple

Employee

Eid	Ename	dept-it
E1	A	1
E2	B	1
E3	B	2

{ t | t ∈ Employee }

it will produce all attributes of the Employee table in the output.

#Q. Retrieve tuples from the relation student where marks are more than 50

Student(sid, name, marks)

Student

Sid	Sname	Marks

Soluⁿ : (i) $\{ t \mid \underbrace{t \in \text{Student}}_{\text{Predicate}} \underbrace{(t.\text{marks} > 50)}_{\text{Condⁿ}} \}$

(ii) $\{ t \mid \exists S \in \text{Student} (S.\text{marks} > 50 \wedge$ $t.\text{sid} = S.\text{Sid} \wedge t.\text{sname} = S.\text{Sname} \wedge t.\text{marks} = S.\text{Marks})$

Creation of attributes w.r.t output tuple t.

#Q. Retrieve sids of all the students from the relation student whose marks are more than 50
Student(sid, name, marks)

Soluⁿ. (i) $\{ t.sid \mid \underbrace{t \in Student \ (t.marks > 50)}_{\text{Predicate Condⁿ}} \}$

(ii) $\{ t \mid \underbrace{\exists S \in Student \ (\underbrace{t.sid = S.sid}_{\text{creation of attribute}} \ \wedge \ S.marks > 50)}_{\text{Predicate Condⁿ}} \}$

$\wedge$: AND

#Q. Retrieve sids and names (Sname) of all the students from the relation student whose marks are more than 50 .
Student(sid, name, marks)

Soluⁿ.

(i) $\{ t.sid, t.sname \mid t \in Student \ (t.marks > 50) \}$

attribute Sid & Sname are required from the selected tuples

(ii) $\{ t \mid \exists S \in Student \ (S.mark > 50 \ \wedge \ t.Sid = S.Sid \wedge t.sname = S.sname) \}$

Creation of attribute corresponding to selected tuple 'S'

#Q. Consider the following relational schema:

Suppliers(sid, sname, city)

Parts(pid, pname, color)

Catalog(sid, pid, cost)

Find sid of all the suppliers who have supplied part with pid=P1

Soluⁿ: (i) $\{ t.sid \mid t \in Catalog \ (t.pid = P_1) \}$

(ii) $\{ t \mid \exists C \in Catalog \ (C.Pid = P_1 \ \wedge \ t.Sid = C.Sid) \}$

#Q. Consider the following relational schema:

Suppliers(sid, sname, city)
Parts(pid, pname, color)
Catalog(sid, pid, cost)

Find sids of all the suppliers who have supplied some 'red' color parts

Soluⁿ (i):

$$\Rightarrow \{t \mid \exists C \in Catalog\ (t.Sid = C.Sid \wedge \exists P \in Parts\ (C.Pid = P.Pid \wedge P.Color = 'RED'))\}$$

(or)

$$\Rightarrow \{t \mid \exists C \in Catalog\ (\exists P \in parts\ (C.Pid = P.Pid \wedge P.Color = 'RED') \wedge t.Sid = C.Sid)\}$$

(ii) $$\{t.Sid \mid t \in Catalog\ (\exists P \in Parts\ (t.Pid = P.Pid \wedge P.Color = 'RED'))\}$$

#Q. Consider the following relational schema:

Suppliers(sid, sname, city)

Parts(pid, pname, color)

Catalog(sid, pid, cost)

Find sids of all the suppliers who have supplied some 'red' color parts or some green Color Parts.

Soln:

(i) $\{t \mid \exists C \in Catalog (t.Sid = C.Sid \wedge \exists P \in Parts (P.Pid = C.Pid \wedge (P.Color = 'RED' \vee P.Color = 'Green')))\}$

(ii) $\{t.Sid \mid t \in Catalog (\exists P \in Parts (t.Pid = P.Pid \wedge P.Color = 'RED'))\}$

$\{U.Sid \mid U \in Catalog (\exists P \in Parts \vee (U.Pid = P.Pid \wedge P.Color = 'Green'))\}$

#Q. Consider the following relational schema:

Suppliers(sid, sname, city)

Parts(pid, pname, color)

Catalog(sid, pid, cost)

Find sids of all the suppliers who have supplied some 'red' color parts and some green Color Parts.

(i) $\{t \mid \exists C_1 \in Catalog (\exists P_1 \in Parts (C_1.Pid = P_1.Pid \wedge P_1.Color = 'RED') \wedge \exists C_2 \in Catalog (\exists P_2 \in Parts (C_1.Sid = C_2.Sid \wedge C_2.Pid = P_2.Pid \wedge P_2.Color = 'Green') \wedge t.Sid = C_1.Sid))\}$

(ii) $\{t.Sid \mid t \in Catalog (\exists P \in Parts (t.Pid = P.Pid \wedge P.Color = 'RED'))\}$

$\wedge$

$\{u.Sid \mid u \in Catalog (\exists P \in Parts (u.Pid = P.Pid \wedge P.Color = 'Green'))\}$



2 mins Summary

Topic Tuple relational calculus

Slide

THANK - YOU

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